



United States Department of the Interior

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October 31, 2014

Memorandum

To: Anne Marie LaRosa, Refuge Manager

From: McCrea Cobb, Wildlife Biologist

Subject: Results of 2014 aerial survey of mountain goat on Kodiak Island, Alaska.

SUMMARY

Biologists from the Kodiak National Wildlife Refuge and Alaska Department of Fish and Game (ADF&G) surveyed mountain goat on Kodiak Island in August and September 2014. We counted a total of 2,513 goats, which included 1,956 adults and 557 kids (28 kids: 100 adults). After adjusting our raw counts for estimated goat sightability (0.68, SE = 0.02), we estimated a total of $3,696 \pm 205$ (95% CI) goats in the survey region. Adequate weather conditions during the 8 survey days (overcast to sunny with wind speeds <15 kts) allowed us to survey alpine habitats in 7 of 8 ADF&G mountain goat hunt areas on Kodiak Island. Counts were higher than 2013 in hunt areas 471 (Wild Creek/Center Mountain, +70%), 473 (Hidden Basin/Terror, +21%), and 478 (south road system, +10%), and lower in adjacent southern hunt area 472 (Crown Mountain, -47%). Some of these changes in abundances may be due to goat movements between adjacent hunt areas. Overall, we counted a record number of goats in hunt area 480 (Spiridon River southward) (1,565 goats) and 22% more (286) than counted in 2013. Counts in the northern portion of hunt area 480 (formerly hunt area 475) were the highest recorded to date (751 goats). We also counted record high number of goats in the southern portion of 480 (former hunt area 477) (813 goats). Productivity (kids: 100 adults) varied across Kodiak and was highest in hunt areas 471 (Wild Creek/Center Mountain, 35 kids: 100 adults), 474 (Uganik River, 39 kids: 100 adults), and 478 (south road system, 36 kids: 100 adults). Productivity was lowest in hunt areas 472 (Crown Mountain, 13 kids: 100 adults) and 473 (Hidden Basin/Terror, 24 kids: adults), which encompasses the initial introduction site. Lower productivity in this region may be indicative of lower habitat quality. Although not included in survey results, we opportunistically observed 25 goats (13 adults and 12 kids) at Cape Grant (north side of Halibut Bay). This group was approximately 35 km from the nearest observed goat group and occupied non-typical habitat compared to other goat sub-populations of Kodiak Island. We recommend that Refuge and ADF&G biologists continue in their efforts to develop a more cost-effective and

logistically-feasible mountain goat survey method and continue to take steps toward developing empirically-derived estimates of nutritional carrying capacity to base goat harvest objectives.

BACKGROUND

Eighteen mountain goats (7 males and 11 females) were introduced to the Hidden Basin region of Kodiak Island from the Kenai Peninsula, in 1952 and 1953. By 1964, the population expanded to 26 goats. The first hunting season was authorized in 1968 when the estimated population was 71 goats. To promote population growth, goats were lightly harvested initially using a restrictive draw hunt within a limited area. The population increased through the 1980s from 150 to over 400 goats. In response, permits changed from a draw hunt to a registration hunt in 1984 and 1985. However, this change resulted in high hunter densities, less harvest selectivity, “herd shooting”, and wanton waste (Smith 1986). Therefore, drawing system regulations were reestablished the following year and remained in place through the 1990s. The population continued to grow in the 1990s, and ADF&G responded by opening new hunt areas: hunting permits were issued for the first time in the southern portion of Kodiak Island in 1991, and a new hunt area close to the road system was opened in 1995. By 1999, the population increased to an estimated 1,000 goats, and the population was believed to occupy all available habitats on the island. By 2001, all of Kodiak and Uganik Islands were open to goat hunting, and a new hunt area in the northern portion of the road system was opened.

In 2001, the Kodiak-Aleutian Regional Advisory Council (KARAC) received a proposal to designate mountain goat as a species eligible for harvest under federal subsistence regulations. In response, a joint (Kodiak Fish and Game Advisory Committee and KARAC) working group formed to discuss ways to meet subsistence needs of rural residents while retaining state harvest management. The Board of Game approved a proposal from the working group to allow village-based registration hunts following conclusion of the drawing hunt season, if allowable surpluses of goats exist. This regulatory change halted potential designation of mountain goat as a federal subsistence resource on Kodiak Island. In 2009, the state of Alaska approved more aggressive harvest management measures (registration hunt) in a segment of the population receiving lower hunting pressure (hunt area 480, south of Spiridon River) because of limited floatplane accessibility. The goat population continued to expand, and by 2011, grew to an estimated population size of 2,400-2,600.

In 2012, a subcommittee of the Alaska Board of Game’s Kodiak Advisory Committee proposed changes to Kodiak’s mountain goat harvest regulations in hunt area 480 in response to the continued rapid growth of the goat population in central and southern portions of the island. The subcommittee included members of the Subsistence Advisory Council, the Board’s Advisory Committee, ADF&G and Refuge biologists, and interested public. The subcommittee generated a proposal for regulation changes which were adopted by the Advisory Committee, endorsed by Kodiak-based state and federal fish and wildlife managers, and submitted for consideration by the Alaska Board of Game. During their winter 2013 meeting, the Board approved the proposal with modification. The revised regulation increased the annual bag limit in RG480 from one to two goats and extended the season closing date from 20 December to 20 March. These regulatory changes were implemented on 1 July 2013.

Surveys of mountain goat abundance have been conducted annually by the ADF&G since 1975 to guide harvest management. Kodiak Refuge began assisting with surveys in 2005. Although survey effort has varied annually, from island-wide counts to partial surveys, results provide baseline data on historical goat population growth, kid recruitment, and range expansion. However, the logistics of surveying goats has become increasingly complex as the population has expanded in range and abundance. Marginal flying conditions and limited budgets have been the primary reasons that island-wide surveys have been irregular. Aerial surveys require high cloud ceilings and low wind speeds, which infrequently occur in mountainous regions of Kodiak Island. Given these challenges, and the potential for further population increases, a more statistically robust and logistically feasible method to quantify goat trends or abundance is needed. When combined with data on mortality rates and productivity, an empirically-derived estimate of abundance could improve our understanding of the relationship between harvest and goat abundance, which would allow managers to achieve the goal of a long-term sustainable goat population that does not negatively impact alpine habitats.

STUDY AREA AND METHODS

Surveys were conducted from a fixed-wing tandem aircraft (CubCrafters Top Cub) piloted by a Refuge pilot (Kurt Reese or Kevin Van Hatten) and carrying Refuge wildlife biologist (McCrea Cobb or William Leacock) as the passenger (hunt area 480), or from a similar fixed-wing tandem aircraft (Cessna O-1 Bird Dog) piloted by a charter pilot (Doug Hilty) and carrying an ADF&G wildlife biologist (John Crye) as the passenger (other hunt areas). Both pilots and passengers acted as observers.

We surveyed in alpine habitats on Kodiak Island above shrub-line (approximately >300 m above sea level) characterized by alpine landcover set in mountainous relief in established ADF&G mountain goat hunt areas. The Refuge pilot attempted to remain at or around 150 m above ground level while surveying, which was viewed as balance between being able to identify goats and having a wide enough field of view on either side of the aircraft.

Upon encountering goats, observers tallied the size of the group and number of kids. Mountain goats were considered an individual group if they were separated by less than approximately 500 m from other goats and behaviorally responded to the observers in a cohesive manner. The passenger recorded the number of adults and kids, the landcover type, and the topography, and a GPS waypoint of the group on paper datasheets (ADF&G), or on a GPS-enabled laptop computer (Dell Latitude with Garmin GLO GPS) running DNRSurvey, an extension to ArcGIS 10.1 that allows for spatially referenced voice records.

We quantified productivity as the ratio of kids: 100 adults. We considered counts of goats in hunt areas that were entirely surveyed as a measure of minimum population size within these areas. To account for imperfect detection and estimate goat abundance with statistical confidence, we multiplied minimum population sizes by a sightability estimate, defined as the mean and standard error of all known goat sightability estimates from published surveys results

(68%, SE = 0.02) (Houston et al. 1986, Cichowski et al. 1994, Poole et al. 2000, Gonzalez-Voyer et al. 2001, Pauley and Crenshaw 2006, Poole 2007, Rice et al. 2009, Jenkins et al. 2012).

RESULTS

We surveyed for 28.6 hours (22.1 hrs by Kodiak Refuge and 6.5 hrs by ADF&G) over 8 survey days (1, 6, 15, 18, 19, 20, and 21 August and 25 September, 2014) (Figures 1 and 2). We counted 2,513 mountain goats in 328 groups (Figure 3 and 4, Table 1). On Kodiak Refuge, we counted 1,784 goats (71% of total) in 253 groups. The median group size was 3 goats and the mean was 8 goats (SE = 0.7). Group sizes ranged from 1 to 71 goats. After adjusting our raw counts for estimated goat sightability, we estimated $3,696 \pm 205$ (95% CI) observable goats in the survey region.

We counted 557 kids, which is a record high. Productivity (kids: 100 adults) was higher (28 kids: 100 adults) than the previous 10-year average (2004-2013, 23 kids: 100 adults) and the same as the long-term average (28), despite higher goat densities this year.

Productivity varied spatially among hunt areas from 13 kids: 100 adults at Crown Mountain (hunt area 472) to 39 kids: 100 adults at Uganik (hunt area 474). The apparent low productivity at Crown Mountain compared to 2013 (25 kids: 100 adults) could be a result of nanny group movement from this region to adjacent areas such as Wild Creek/Center Mountain (hunt area 471), where productivity increased from 7 kids: 100 adults in 2013 to 35 kids: 100 adults. We observed the opposite pattern in 2013 (decreased productivity at Wild Creek/Center Mountain and increased productivity at Crown Mountain), suggesting interannual movement among these hunt areas.

Within hunt area 480, which encompasses most of the Refuge lands on Kodiak Island, we counted a record high number of goats (1,564). Despite high densities of goats, productivity remained unchanged from 2013 (28: 100 kids) in this region, suggesting sustained growth. Within hunt area 480, we observed more goats in the northern Uyak/Spiridon region (formerly 475) and more goats in the southern Karluk/Kaiagnak region (formerly 477) than have ever been previously recorded.

Even though we observed a record number of goats, our count does not represent an island-wide “census” because we did not survey all available mountain goat habitats. Most notably, we did not survey hunt area 476, where 84 goats were counted in 2013. We observed 25 goats (13 adults and 12 kids) on Cape Grant (north end of Halibut Bay) during sea otter surveys in July 2014, but these individuals were not included in the total count for 480 because we did not survey this region during the August/September goat surveys. The Cape Grant group was separated from the nearest goat group observed during surveys (Fraser/Red Lake region) by approximately 35 km of rolling mountain and flat tundra valley habitats.

DISCUSSION

Kodiak's mountain goat population has exhibited exponential-like growth since their introduction (Cobb 2011). However, survey results indicate that population growth across different regions of Kodiak has varied. Areas near the initial introduction site (hunt area 471, 472, and 473) appear to have peaked in abundance in the 1980s and 1990s, and subsequently have stabilized at lower abundances. Along the road system (hunt areas 478 and 479), the population rapidly increased in the mid-1990s and, since that time, has remained relatively stable. Within Kodiak Refuge (hunt areas 480 and 474, and portions of 473 and 476), the populations exhibited rapid growth over the past 10 years. Currently, 71% of the surveyed goat population occurs within the Refuge.

Continued increases in goat abundances have prompted concerns by the Kodiak Refuge over potential impacts to alpine flora and fauna. High densities of ungulates can cause adverse ecological impacts (Hobbs 1996), but the relationship between Kodiak goat densities and their effects on vegetation is unclear. Introduced ungulate populations that are free from predation and have abundant forage often follow a general trend of rapidly increasing to peak abundances, crashing to low levels, and then increasing to a theoretical carrying capacity below peak abundance (Caughley 1970). During the initial growth phase, ungulates can cause detrimental landscape-level effects by altering vegetation structure and composition, soil system functioning, and chemical processes (Hobbs 1996, Spear and Chown 2009). Empirically-driven harvest management can successfully slow irruptive population growth rates, resulting in less natural resource damage and an elevated long-term ecological carrying capacity (i.e. more harvest opportunities in the long-term). Hunter harvests offer a potential tool in regulating Kodiak's mountain goat population sizes, but currently the relationship between harvests and annual changes in population abundance is unclear. Quantifying the effects of hunter harvest on annual changes in population size is needed to maintain a viable mountain goat population that limits impacts to Refuge resources while providing an opportunity for hunter harvest.

Our results suggest that mountain goat abundances in the northern and southern portions of Kodiak continue to increase, and conversely, abundances in the central portion of the island are stable or declining. Between 2013 and 2014, counts in northern hunt areas (471, 478 and 479) increased 22% from 538 to 655 goats. Some of this increase in northern Kodiak may be attributable to an apparent increase in kid abundances; kid counts increased from 133 kids in 2013 to 160 kids this year. Increases observed in northern hunt areas may also be partly due to movement from adjacent hunt areas, namely hunt area 472, where abundances declined from 25 to 17 goats, between 2013 and 2014. Alternatively, we speculate that increases in southern Kodiak (480) are likely due to high recruitment and low mortality rates. Goats generally exhibit differences in habitat use across seasons through elevational movements that are driven by changes in the spatial distributions of available high-quality forage. At Kodiak, goat movement patterns remain unclear, but results from an ongoing study of GPS-collared goats may lead to a better understand.

Mountain goat productivity on Kodiak (28 kids: 100 adults) was higher than more well-established native mountain goat populations in coastal Alaska. At Kenai Peninsula, where goat populations have declined 30-50% from the early 1990s, productivity averaged only 18 kids:

adult between 2009 and 2011 (McDonough 2012). In the Ketchikan region (ADF&G Game Management Unit 1A), productivity averaged 22 kids: 100 adults between 2000 and 2010 (Porter 2012). Mountain goat populations in the eastern Gulf of Alaska coast (ADF&G GMU 5) averaged 22 kids: 100 adults between 2005 and 2008 (Lowell 2008). Higher productivity at Kodiak may indicate better habitat quality compared to other regions in Alaska. This conclusion is further supported a study that found early horn growth, an indicator of habitat quality, was greater in Kodiak goats than in Kenai Peninsula goats (McDonough et al. 2006).

Ungulate productivity in populations experiencing little predation pressure, such as goats on Kodiak Island, is often a function of high-quality forage availability. The long-term survey results of goats from Kodiak Island indicate that productivity varies among hunt areas, and this spatial variation may relate to herds' proximity to the initial introduction site. Results from the past 10 years (2004-2013) show that hunt areas near the introduction site (471, 472, 473, and 474) have lower productivity (22 kids: 100 adults) compared to hunt areas further from the introduction site during the same time period, such as those on the road system (478 and 479) and in southern portions of Kodiak (480) that averaged (27 kids: 100 adults).

Documentation of a rapidly growing goat population in hunt area 480 prompted recent regulatory changes aimed at increasing hunter harvest, slowing population growth, and minimizing the potential for habitat degradation. In 2010, a registration hunt opened in hunt area 480. Despite this change, goat harvests did not increase, and the population continued to grow exponentially. Annual bag limits were increased during the 2013 season to two goats, and the season closure date was extended from December to March, in hunt area 480. In response, harvests increased in hunt area 480 between the 2012 and 2013 seasons from 82 to 175 goats killed. The number of hunters in the field in hunt area also increased from 221 in 2012, to 446 in 2013. At this point, it is unclear which of the two regulatory changes had the most impact on harvest levels. Despite a doubling in harvest and hunters in the field in hunt area 480, goat counts in this region increased during the same time period by 22% (1,278 to 1,564 goats). It is still too early to determine whether longer term population growth has been affected by the regulatory changes. We expect more clarity regarding this determination after two to three seasons under the new regulations.

In general, probable barriers to maximizing hunter harvest pressure in hunt area 480 include:

- The remoteness and associated high costs of air transportation compared to other goat hunt areas on Kodiak. Rising aviation fuel costs may further limit the number of hunters willing to cover transportation costs to hunt area 480.
- Additional costs associated with guided hunting services that are required of non-resident hunters.
- The current restriction on same-day airborne hunting of goats.
- The limited number of floatplane-accessible alpine lakes.

If further harvest pressure in hunt area 480 is needed, the following changes are recommended:

- Allow non-residents to hunt 480 without a guide
- Allow same-day airborne hunting of goats in 480

- Increase hunter access by encouraging wheeled aircraft landings on alpine meadows and ridge tops. Provide public outreach to air charters and the public on the legality of wheeled landings.
- Remove “billy harvest preferred” admonition in the state harvest regulation booklet. Replace with “nanny harvest preferred”.
- Subdivide hunt area 480 into smaller and more manageable hunt areas based on current goat densities and ease of hunter access. Harvest pressure and access varies widely across 480, which covers almost half of Kodiak Island. Smaller hunt areas within 480 would allow for liberalizing harvest pressure in areas of high goat density (Uyak/Spiridon) while maintaining populations in areas of lower density that are more easily accessible (Hepburn/Frazer).
- Extend the hunting season in 480 from March 20 to mid-April.

Our estimate of observable goat abundance in the survey area, $3,696 \pm 205$ (95% CI), is a continuation of a first effort toward developing a statistically-robust estimate of the mountain goat population that accounts for sightability bias. Because goat sightability has not been locally estimated, we employed an average goat sightability estimate to the survey counts based on studies from British Columbia (Cichowski et al. 1994, Poole et al. 2000), coastal Washington (Houston et al. 1986, Rice et al. 2009, Jenkins et al. 2012), Idaho (Pauley and Crenshaw 2006), and Alberta (Gonzalez-Voyer et al. 2001). Goat sightability estimates in the literature vary from 59% in Idaho (Pauley and Crenshaw 2006) to 78% on the Olympic Peninsula, Washington (Rice et al. 2009). On Kodiak Island, factors that likely influenced sightability included variation in observer experience, snow cover, lighting, and color of terrain (Rice et al. 2009). Goats were more difficult to spot on snow, and on granite compared to other substrates. Granite cliffs are often associated with smaller boulders, which further limited observer sightability. Granitic habitats were common in the northern portion of hunt area 480, and hunt areas 476 and 474. Consequently goat sightability was likely lower in areas of granitic terrain compared to non-granitic terrain, but this was not reflected in the current population estimate. Unfortunately, the current survey method does not account for any of these factors. Because surveys were restricted to alpine terrain, mountain goats occupying lower elevation habitat and outside of the survey range were excluded. The abundance of this segment of the population was unknown but assumed small (>100 goats).

Conducting island-wide population counts of Kodiak Island goats has become increasingly costly because of the increasing size and range of the goat population. Completing island-wide counts is also logistically difficult because of the unpredictable weather. An improved sampling design is needed to reduce survey costs while yielding accurate and precise parameter estimates. To improve survey results, we recommend that future efforts focus on developing a quantitative estimate of goat sightability for Kodiak Island. Quantifying sightability during wildlife surveys has received much attention recently, and field and statistical methods for determining sightability and confidence intervals have greatly improved since these surveys were initiated (Seber 1992). In southeast Alaska, ADF&G biologists developed a mark-recapture method for estimating goat sightability using collared goats (White 2010). This method might also be applicable to Kodiak Island. As a first step toward applying this method to Kodiak, ADF&G biologists captured and fitted 15 goats with GPS collars in June 2013 and they plan to deploy additional collars in 2015.

The introduced mountain goats of Kodiak Island present a unique hunting opportunity that has accounted for 32% of Alaska's annual goat harvest (Van Daele and Crye 2012). However, if the population continues to grow, it also presents a potential threat to native flora and fauna, as well as hunter harvest opportunity. Goats now likely occupy most available habitats and the population size is likely more than tripled ADF&G management goal of 1,000 goats. Empirically-based harvest management is critically needed to maintain a productive goat population that meets hunters' expectations while maintaining habitat quality. It is important that Kodiak Refuge and ADF&G work together toward meeting management goals, including a cost-effective survey technique that estimate population size and an empirical carrying capacity estimate to direct harvest management.

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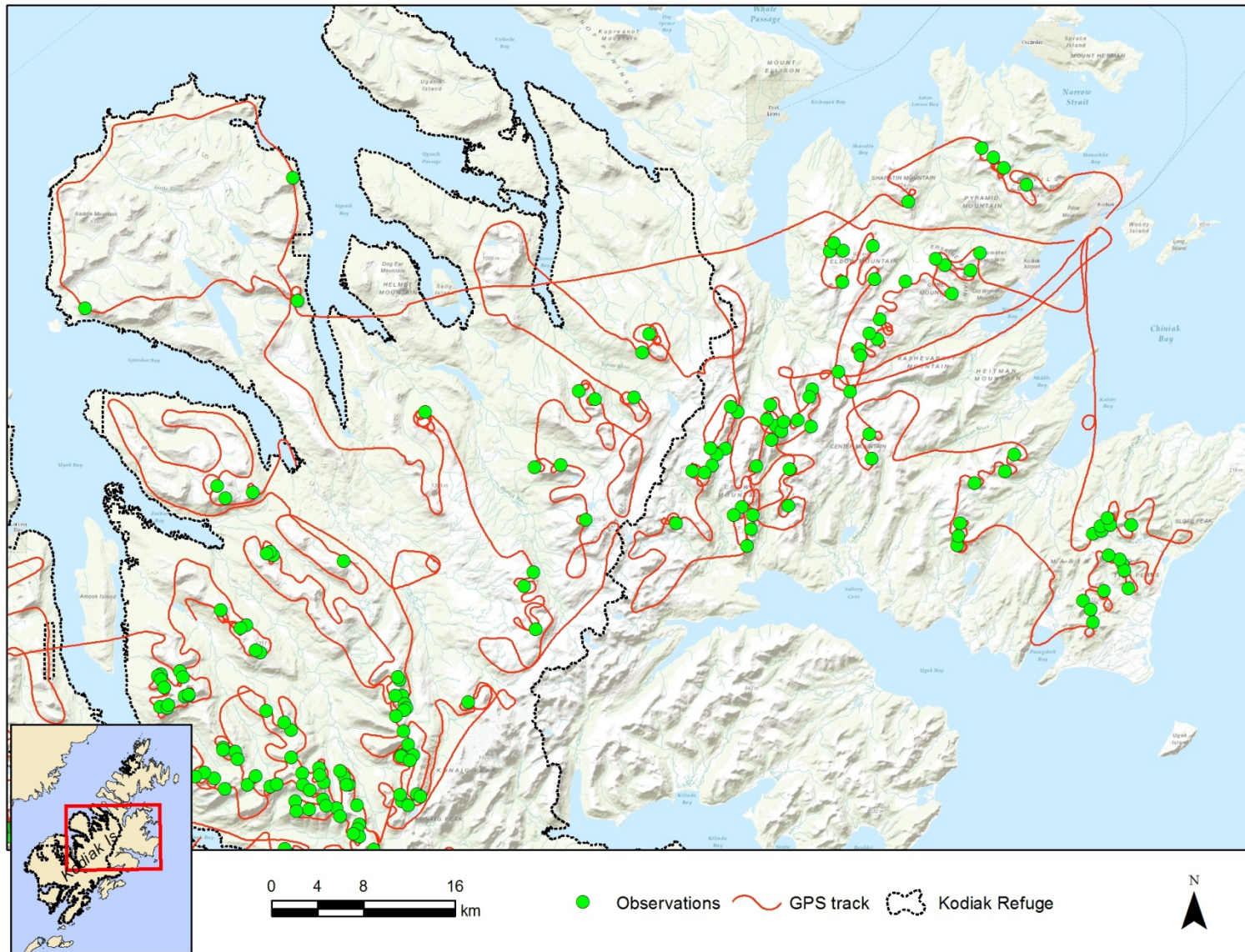


Figure 1. Tracks of survey aircraft and locations of mountain goat groups during aerial surveys, northern Kodiak Island, 2014.

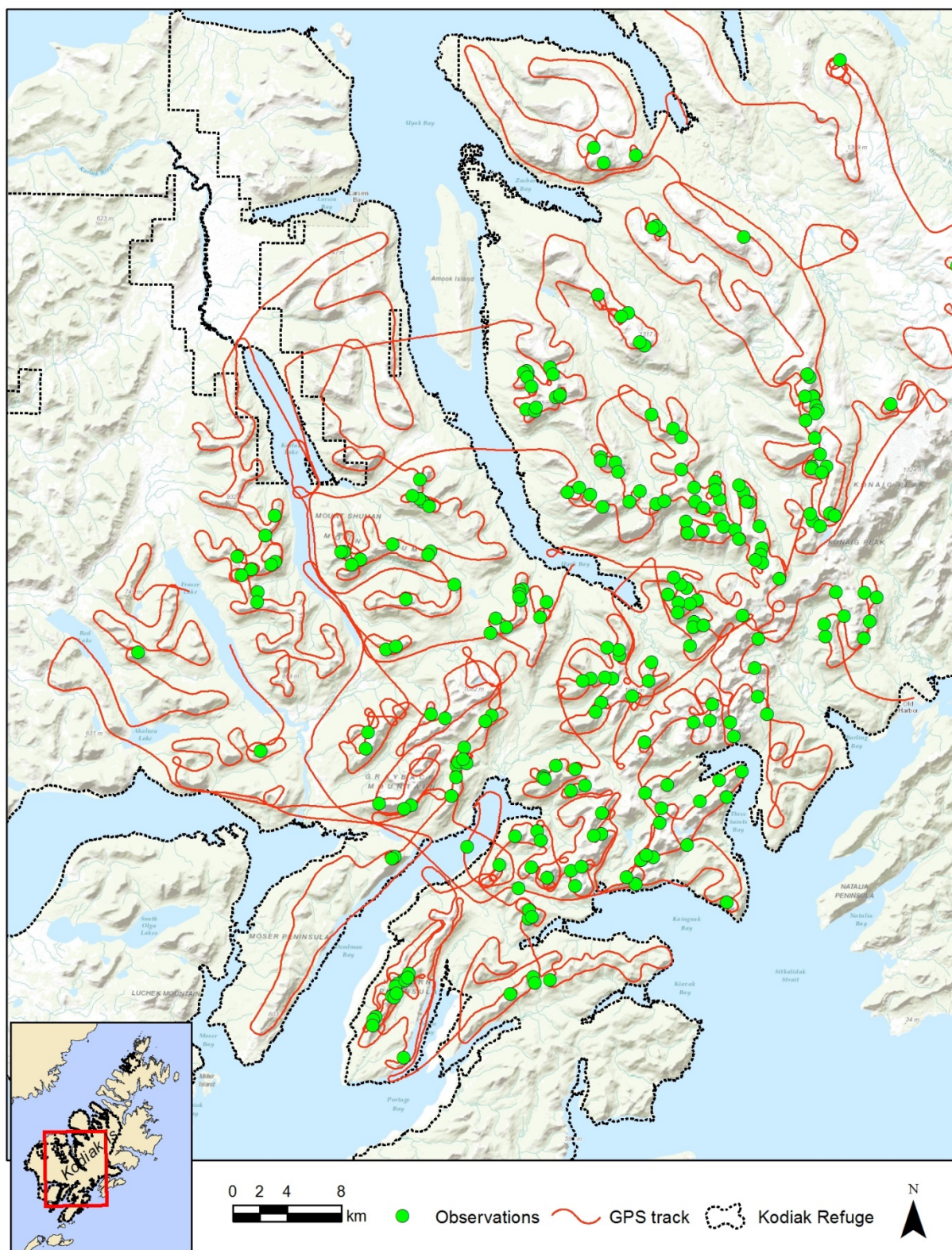


Figure 2. Tracks of survey aircraft and locations of mountain goat groups during aerial surveys, southern Kodiak Island, 2014.

Table 1. Summary of mountain goat survey results, by hunt area, Kodiak Island, Alaska, 2014.

Hunt area	Description	Total	Adults	Kids	% kids	Kids: 100 adults
471	Wild Cr./Center Mt.	101	75	26	26	35
472	Crown Mt.	17	15	2	12	13
473	Hidden Basin/Terror	61	49	12	20	24
474	Uganik	215	155	60	28	39
476 ¹	Kiliuda	--	--	--	--	--
478	South road system	335	246	89	27	36
479	North road system	219	174	45	21	26
480	Spiridon southward	1,565	1,242	323	21	26
Total		2,513	1,956	557	22	28

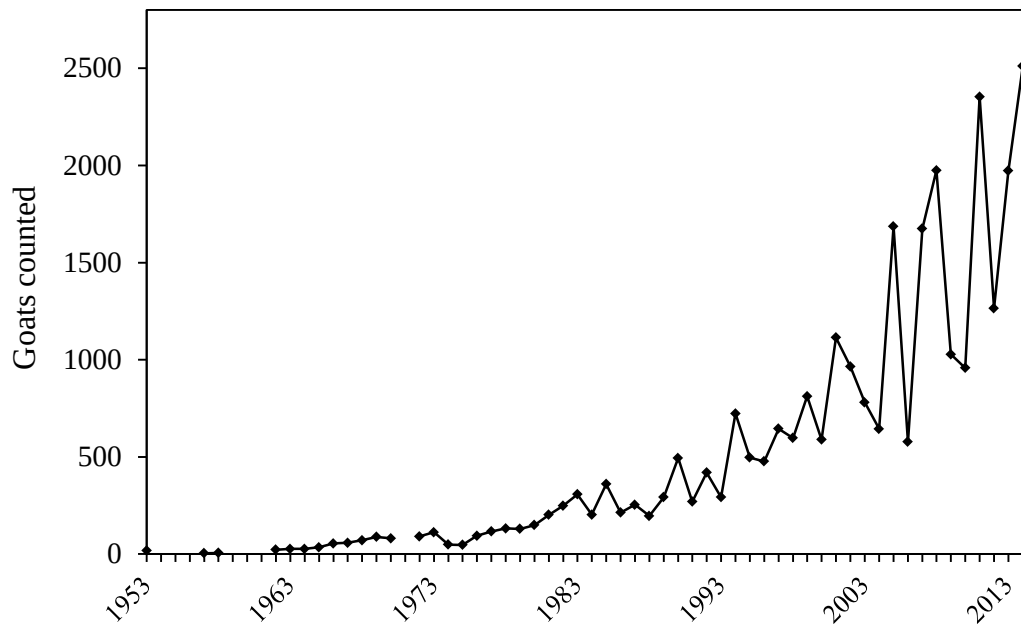
¹ Not surveyed

Table 2. Summary of mountain goat survey results from the past 10 years (2005-2014), Kodiak Island, Alaska. Data were collected by ADF&G and Kodiak Refuge biologists. Survey effort varied annually. Island-wide surveys were conducted in 2005, 2008, and 2011.

Year	Total	Kids: 100 adults
2005	1,686 ¹	23
2006	577	22
2007	1,674	20
2008	1,975 ¹	23
2009	1,028	26
2010	950	18
2011	2,364 ¹	20
2012	1,265	18
2013	1,973	28
2014	2,513	28

¹ Considered island-wide count and minimum estimate of goat abundances on Kodiak

A)



B)

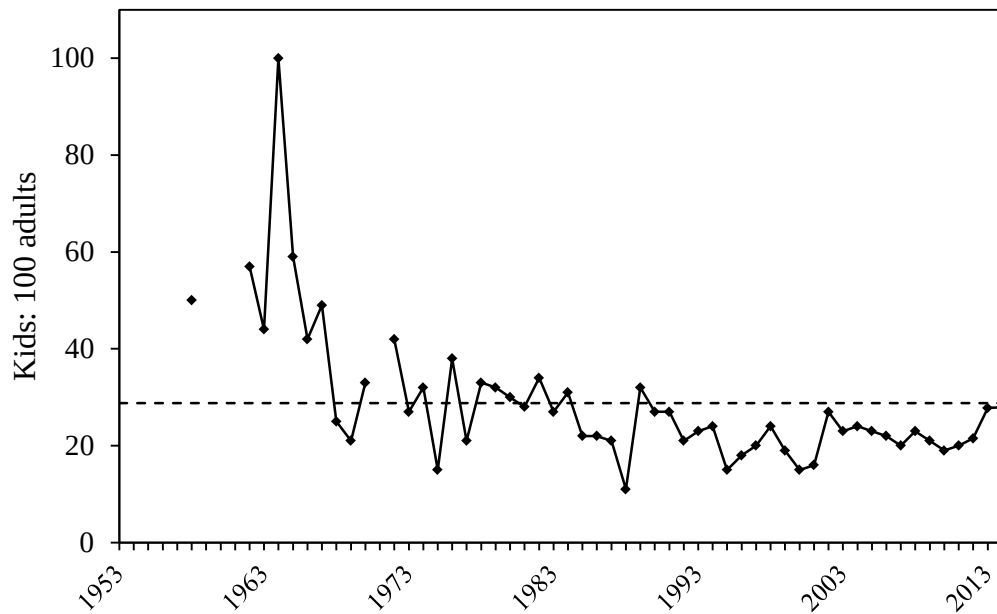
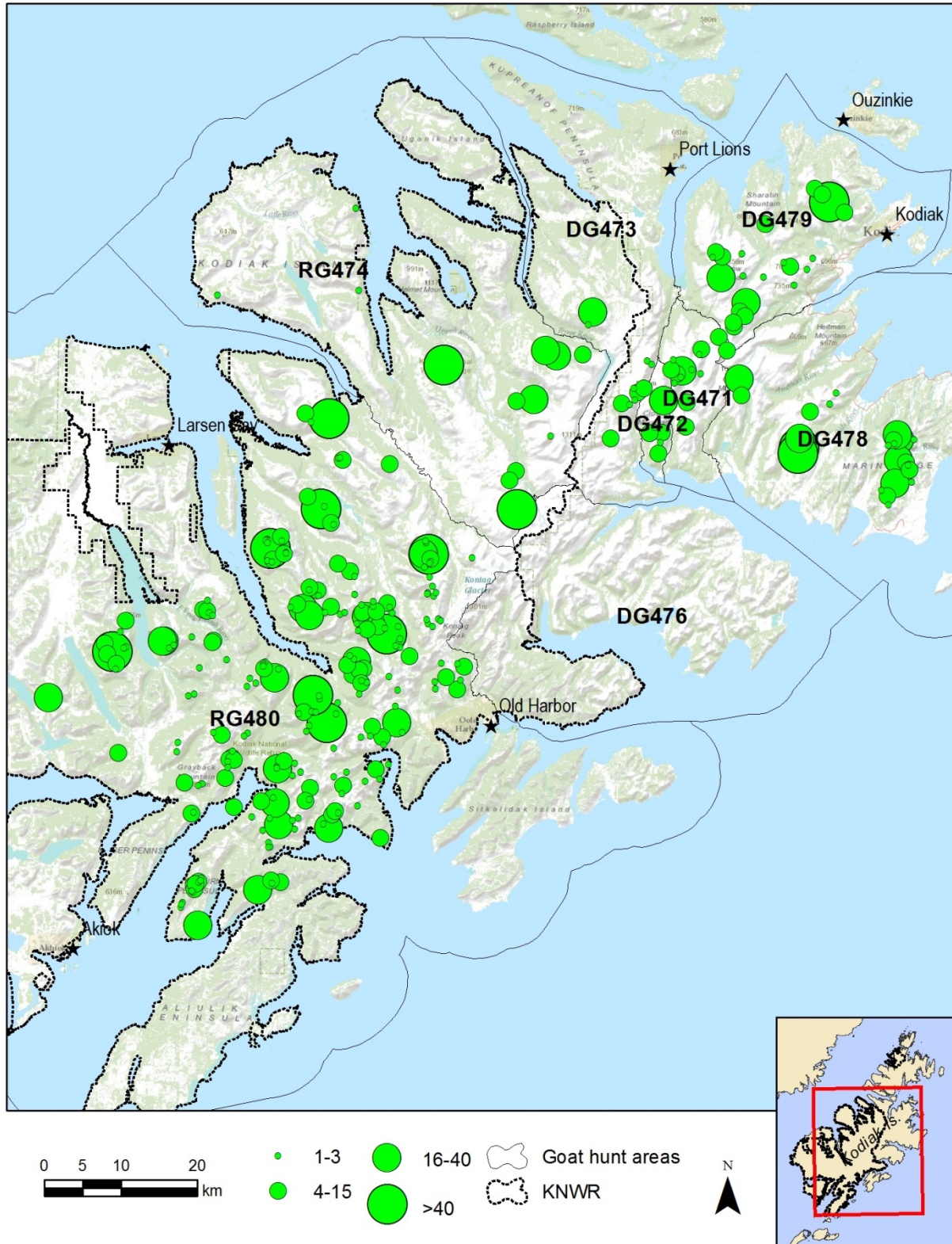


Figure 3. A) Number of mountain goats and B) kids: 100 adults observed during summer mountain goat aerial surveys, Kodiak Island, Alaska, 1953-2014. Survey effort varied annually and not all survey years were island-wide counts. The dotted line in B) indicates the average productivity, 1953-2014 (28 kids: 100 adults).



Appendix A. Number of mountain goats counted, by hunt area, between 1995 and 2014. Not all hunt areas were surveyed every year.

